

1. Data is raw, unprocessed facts or figures that hold no individual meaning on their own. When data is labeled and given context, then it is given meaning and thus turns into information.

2. Metadata is essentially data that describes data. When data is turned into information, metadata is used to describe the data to turn it into information.

3. DBMS stands for database management system, and it is a system that acts as an intermediate between the user and the database. The DBMS manages data on both sides, maintaining the database structure and controlling access to the data in the database.

The advantages are that security is ensured because the DBMS protects the data in the database by managing access to the data. Another advantage is that the DBMS is organized, so data access is cleaner and more centralized.

4. Operational databases (or OLTP - online transaction processing) stores current data that is used for daily operations. These databases are often optimized for speed and efficiency.

Analytical databases (or OLAP - online analytical processing) stores historical and aggregated data, used for more complex projects in data analysis. These databases are often optimized for queries and reporting.

They are both similar because they are both still databases that store data which is accessed by the user. They are designed for clean, organized listing of data, while protecting it at the same time.

They are different in operation and purpose. Operational databases are meant for speed and smaller pieces of data, whereas analytical databases are meant for more complex and bigger amounts of data.

5. NoSQL has a wide variety of use cases because of its wide capability of storing different kinds of data. NoSQL can be used for general-purpose databases, large volumes of data with simple lookup queries, large amounts of data and predictable query patterns, and traversal among relationships between data.

6. SQLite does not require server configuration. The advantages of SQLite are that it is lightweight, reliable, and fast. Because of its self-contained nature, SQLite can be used for things like mobile applications, embedded systems, and other small-scale projects.

7. ACID stands for atomicity, consistency, isolation, and durability. ACID properties ensure that DBMSs have data integrity and consistency.

Atomicity is when data can no longer be divided into smaller units.

Consistency is the accuracy of the database.

Isolation means parallel activity should not affect each other.

Durability describes the permanence of the data, remaining unchanged even with crashes.